



Smart Drain Flood Monitoring System

A Real-Time IoT-Based Early Warning System

Presented by:

Aditya Mathad

1RN24EC008

Christo Savio

1RN24CD01

George

9

Sandeep A Patil

1RN24EC117

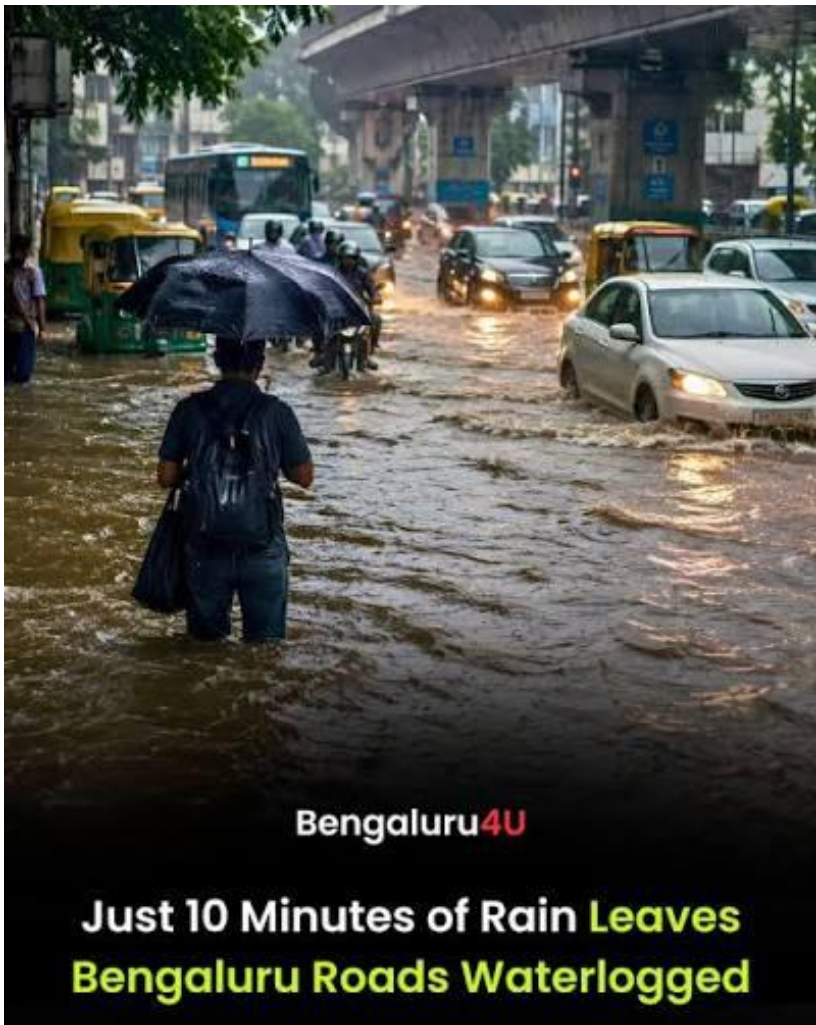
Vivek Patne M

1RN24CD112

THE PROBLEM

Heavy Rain, Flooded Roads, No Warning

- A few hours of heavy rain is enough to flood roads, drains, and low-lying areas across the city.
- Waterlogging disrupts traffic, damages homes and vehicles, and puts lives at risk every monsoon.
- Drains are usually checked only after flooding has already happened.
- Citizens have no easy way to warn each other or alert authorities in real time.



NOT DESIGNED FOR THIS, DON'T CLOG DRAINS: BBMP

60% Of B'uru saw heavy rain, moderate rain in remaining areas	2nd Highest rainfall in a decade	33,000 Distress calls to Bescom	20+ Lakes in B'uru overflow	137 Cars, bikes damaged	9.9km Electronics City Elevated E-way shut for hours	50 E-buses suspended by BMTC
---	--	---	---------------------------------------	-----------------------------------	--	--

A tractor being used to rescue people from a waterlogged area in Sal Layout; (L) Flooded Central Silk Board Metro station entrance

WORST-HIT AREAS: Koramangala, Basavanagudi, BTM Layout, Marathahalli, Yelahanka, HAL Airport, Shantinagar, Kengeri, Kodigehalli, Madanayakanahalli, Madavara, Vaderahalli, Chikkabanavara, Chowdeshwarinagar & Kempegowda ward

“...Let's be clear: The issues we face today have been ignored for years, across govts and administrations... we are working to solve them. Not with temporary fixes, but with long-term, sustainable solutions...”

DK Shivakumar | DEPUTY CM & BENGALURU DEVELOPMENT MINISTER

“The infrastructure isn't designed for this (kind of rain). Changing weather and unplanned construction are real challenges. I urge residents not to clog drains”

Maheshwar Rao | BBMP CHIEF COMMISSIONER

Intelligent Water Clogging and Flood Prevention Using Real-Time Data

A smart early-warning system that watches water levels as they rise, predicts flooding before it happens, and alerts people in time to act.



Real-Time Monitoring

Water levels are tracked continuously, with no manual checking needed.



Early Prediction

Live readings plus weather forecasts flag rising risk before it turns critical.



Instant Alerts

Alerts reach a live dashboard, on-site lights, and email the moment risk rises.



Citizen Reporting

Anyone nearby can report waterlogging straight from their phone.

What We Built — and Why It's Worth Adopting

What We Achieved

-  Tracks water levels and classifies risk into clear stages - Normal, Watch, Alert, Critical.
-  Sends alerts well before water reaches a dangerous level, giving real time to react.
-  A live dashboard shows current status, trends, and history at a glance.
-  A citizen-reporting channel is live and tested end-to-end.

Why It's Worth Adopting

-  **Faster Response**
Know before the road is already flooded, not after.
-  **Safer for Everyone**
Fewer surprises for commuters, property, and response teams.
-  **Built to Scale**
Grows from one drain to an entire city without a redesign.
-  **Fits Civic Priorities**
Low-cost and data-backed — easy to plug into control rooms.

How It Works Under the Hood



Real-Time Monitoring

Field data is collected and refreshed continuously.



Data Management

Readings are cleaned, smoothed, and turned into a clear risk status.



Prevention & Control

Automated alerts and controls act before flooding escalates.

BUILT WITH



ESP32 + HC SR04 + Float Switch



MQTT Messaging



Node-RED



Weather Data (OpenWeatherMap)



Node-RED Dashboard



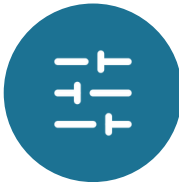
Email Alerts

Where We're Taking This Next



Blockage Detection

Spot clogged drains so maintenance teams know exactly where to act.



Automated Pump / Valve Control

Trigger drainage pumps or valves automatically in high-risk moments.



City-Wide Scaling

Expand from one point to multiple zones, all on one dashboard.



Mobile App & SMS Alerts

Reach citizens directly on their phones, even without internet.



Municipal Integration

Connect into government control rooms and disaster-management systems.